

Dhruva Bhattacharya

Portfolio | [GitHub](#) | [LeetCode](#) | [LinkedIn](#) | dhruvabhattacharya130102@gmail.com

PROFESSIONAL SUMMARY

Backend Software Engineer with production experience designing Java/Spring Boot microservices and REST APIs for large-scale, multi-tenant telecom OSS/BSS platforms, plus hands-on experience building Generative AI/RAG-based agent systems. Cut API latency from **1000ms to 150ms (85% reduction)**, grew transaction throughput by **40%**, and reduced backend processing time by **25%** through caching, query optimization, and API orchestration. Immediate joiner.

TECHNICAL SKILLS

Languages: Java, Python, C/C++ , C#, Bash

Backend & APIs: Spring Boot, Spring MVC, REST APIs, Microservices, API Orchestration, .NET Core, Express.js

Databases & Caching: MySQL, MongoDB, SQL Server, PL/SQL, Data Modeling, Query Optimization

Cloud, DevOps & Tools: Azure, Docker, Kubernetes, CI/CD, Git, Maven, Postman, Swagger/OpenAPI, Azure DevOps

AI/GenAI: RAG Pipelines (Ingestion, Chunking, Embedding, Retrieval), Groq, Llama-3.3-70B, Ollama, Prompt Engineering, Hallucination Mitigation, Copilot Studio, Azure AI Document Intelligence

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS)

Aug 2024 – Present

System Engineer · Backend Developer

TCS HOBS – Telecom OSS/BSS Platform

Java, Spring Boot, Microservices

Multi-tenant platform for global telecom operators (Sonatel, Malaysian Telecom, OGN, Vodafone) – invoicing, billing, ticketing & account management

- Re-architected a slow, multi-join, cross-database query layer into a **consolidated MongoDB document-store cluster**, removing cross-DB joins on high-volume account data and significantly increasing UI data-load throughput.
- Implemented **Redis-backed caching** for high-read endpoints, cutting average API latency from **1000ms to 150ms (85% reduction)** and lifting transaction throughput by **40%**.
- Built **parallel and sequential API orchestration flows** in the account-provisioning layer, cutting backend processing time by **25%** while sustaining **99.9% uptime** under multi-tenant load.
- Delivered a **webhook-driven SLA-breach alert system** that flags and escalates tickets nearing or exceeding a **24-hour SLA** window, and led **migration of legacy SOAP/JAXB (XML) services to a stateless, annotation-driven REST architecture**.
- Root-caused recurring production defects in API mapping, notification dispatch, and vendor integrations, and strengthened reliability via **structured logging, health checks, and smoke tests** – cutting incidents by **18%**, debugging time by **25%**, and defect leakage by **20%**.

Microsoft ESS (Employee Self-Service) AI Agent

Generative AI / Automation

Client: Microsoft, delivered via TCS – built entirely on the Microsoft internal tool stack

- Built **Copilot Studio connectors** to Microsoft-internal APIs, surfacing leave balance, ticket status, payroll, and device-request data inside M365 Copilot.
- Designed **agent-to-agent orchestration workflows** using Azure Logic Apps and .NET Core to route user queries across specialized agents and automate stakeholder notifications.
- Owned **response-quality validation** for agent outputs, benchmarking against expected responses and knowledge-base/system-prompt alignment to consistently clear an **80%+ accuracy** bar before release.
- Built an **Azure DevOps CI/CD pipeline** (Playwright test automation, Key Vault-managed secrets) that aggregates and summarizes responses across ESS sub-agents and publishes them via Power Apps/Logic Apps connectors into Copilot Studio, with Application Insights logging for full observability.
- Delivered **multilingual OCR automation** (Azure AI Document Intelligence) for Purchase Order validation across **6+ languages** (Arabic, Japanese, French, Italian & more), with human-in-the-loop sign-off inside a **1-day SLA** – cutting manual data-entry effort by **78%** for the global finance/BPO team.

INTERNSHIP EXPERIENCE

TechXR Innovations Pvt. Ltd. – AR/VR Intern

Jul 2022 – Sep 2022

- Designed and built a VR-based surgical simulation prototype in Unity, covering environment design, interaction flow, prototyping, deployment, and user-feedback validation.

PROJECTS

Logic Labs – AI-Powered EdTech Platform ([Live Demo](#) | [GitHub](#))

MERN, MongoDB, Groq/Llama-3.3-70B, Razorpay

- Built and hardened a full-stack, role-based EdTech platform (student/instructor/admin) with REST APIs for auth, course management, Razorpay payments, Clouinary media, and progress tracking; shipped a **RAG chatbot** (Groq-hosted Llama-3.3-70B) with an ingestion, chunking, embedding, and retrieval pipeline plus hallucination-mitigating response validation; eliminated N+1 queries and resolved **10+ security vulnerabilities** (bcrypt, CSRF, race-condition prevention).

Production Incident Resolution – OTP Signup Failure

Node.js/Express, MongoDB/Mongoose, Railway, Netlify

- Diagnosed a production signup outage to a Mongoose `post('save')` hook **coupling OTP persistence to email delivery**, with failures silently swallowed by environment-gated logging; **decoupled email sending into the controller layer** with explicit error handling and unconditional production logging.
- Root-caused the failure to Resend's sandbox restriction blocking non-owner delivery; evaluated Gmail SMTP (blocked by Railway's outbound ports) and Resend domain verification (no owned domain) before shipping a **from-scratch Brevo HTTPS API integration**, verified live against owner and non-owner recipients.
- Uncovered a **stale production branch** (4 commits behind main) blocking the fix from shipping; fast-forward merged, verified via direct production API calls, and added **per-provider mail-service health checks** for faster future diagnosis.

EDUCATION

Gyan Ganga Institute of Technology & Sciences

2020 – 2024

B.Tech in Computer Science & Engineering with Specialization in Artificial Intelligence and Machine Learning

CGPA: 8.28/10

CERTIFICATIONS & ACHIEVEMENTS

Certifications: AWS Cloud Practitioner, AWS Cloud Foundations, AWS ML for NLP, Microsoft Azure AI Engineer Associate, **UpGrad Certificate Program in Generative AI**, **Airtribe – AI-First Software Engineering**

Achievements: **Rank 1 (Asia)** – Alibaba Cloud Low-Code Contest (**3000+ teams**); **Smart India Hackathon (SIH) Finalist** – advanced past college and state rounds to the national finale of India's **36-hour** non-stop software hackathon; **900+** LeetCode problems solved